

VARUN SINGHAL

San Diego, CA | (858) 4440091 | varunraisinghal12@gmail.com | linkedin.com/in/varunraisinghal/ | github.com/varunraisinghal/

EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science: Artificial Intelligence

January 2025 – December 2026

Concentration Focus: Artificial Intelligence, Machine Learning, Quantum Computing + Cryptography

University of California, San Diego

San Diego, CA

Bachelor of Science: Computer Science

October 2020 - June 2024

GPA: 3.83/4.00, Provost Honors (2020 - 2024)

PROFESSIONAL EXPERIENCE

GEICO

San Francisco, CA

Software Engineer

May 2024 - August 2025

- Optimized a production quote orchestration and estimation system, reducing **end-to-end latency from ~30s to <13s** and **compute costs by 50%+** while meeting a <15s SLA and preserving quote consistency.
- Re-architected the **critical path** by eliminating unnecessary service calls, **parallelizing independent requests**, reducing duplicate computation, and deferring non-critical processing that did not affect initial pricing decisions.
- Led a cross-team **ML scoring-model initiative** from feature design through production integration to improve premium-estimation accuracy across partner quoting workflows.
- Productionized ML and backend services using **AWS SageMaker, Lambda, DynamoDB, REST APIs, and CI/CD**, improving reliability and supporting high-volume production inference.
- Authored API documentation, technical guides, and video tutorials for accelerated onboarding **across 5+ engineering teams**

Amazon: Amazon Web Services (AWS)

Seattle, WA

Software Engineering Intern

May 2023 - August 2023

- Worked directly with **Fortune 500** customers to design and provision production database and storage pipelines, integrating new workloads with existing enterprise environments and **onboarding 5 customer workloads**.
- Built automated **CloudWatch and GuardDuty** health checks that reduced failure-detection time by **~40%**

RESEARCH & TECHNICAL PROJECTS

VLM4Rec: Multimodal Semantic Representation for Recommendation with Large Vision-Language Models

arXiv, March 2026

Python, PyTorch, LLaVA-NeXT 7B, Sentence-BERT, CLIP, Hugging Face, NVIDIA GPU

- Built an end-to-end **multimodal recommender system** using LLaVA-NeXT 7B semantic descriptions, Sentence-BERT item embeddings, and cosine-similarity retrieval over user histories.
- Designed and optimized a GPU inference pipeline using **4-bit quantized LLaVA-NeXT 7B** on an **NVIDIA RTX 5080**, batching 4,708 catalog items and precomputing/caching semantic embeddings to shift expensive inference offline.
- Evaluated **12 text, vision, and multimodal fusion approaches** across 2,914 users; achieved **0.3539 Recall@10** and **0.2745 NDCG@10**, improving Recall@10 by **54.9%** over the BERT baseline.

Low Latency C++ Arbitrage Engine | C++

- Implemented a graph-based **C++** arbitrage engine and real-time market-data pipeline across Binance, Coinbase, and Kraken with **<10ms detection** and **<300ms execution latency**.

Prediction Market Arbitrage Scanner | Python, FastAPI

- Built a **real-time** pipeline to ingest, normalize, semantically match, and validate Polymarket/Kalshi markets; event-level filtering reduced **LLM calls** by **~80%** while minimizing false-positive matches.

RELEVANT SKILLS

- Languages:** C/C++, Python, SQL, Software Design, Algorithms
- GPU & AI:** CUDA C/C++, CUDA Kernels, Parallel Programming, GPU Programming, NVIDIA GPUs, NVIDIA Nsight, Machine Learning, Deep Learning, AI Algorithms, PyTorch, TensorFlow, Hugging Face Transformers, LLaVA-NeXT, Sentence-BERT, CLIP, Vision-Language Models, Recommender Systems, Computer Vision, NLP
- Systems & Performance:** Linux, Computer Architecture, Low-Level Performance Optimization, Model Inference, Quantization, Information Retrieval, Latency & Throughput Optimization, Performance Profiling, AWS SageMaker, CI/CD